

PDC Multiple-Scattering Ablation: Two-Region Geometry, Three Configurations

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Abstract

This memo records three Geant4 ablation configurations of the SAMURAI + PDC setup that vary the vacuum/air distribution along the proton flight path, in order to quantify the air contribution to the PDC hit-position resolution $\sigma_y(\text{PDC2})$. The target sits inside the dipole yoke cavity (yoke-local coordinate $z = -920$ mm, well inside the ± 1750 mm cavity); the magnet cavity and the `VacuumDownstream` beam pipe are physically connected and share a single `BeamLineVacuum` switch. The flight path is partitioned into **two regions**: Region A = magnet cavity + downstream pipe (target $\rightarrow$ ExitWindow, ~ 4.3 m), Region B = ExitWindow $\rightarrow$ PDC1 $\rightarrow$ PDC2 (~ 2.0 m). The three configurations give $\sigma_y(\text{PDC2})$ of 12.8–14.9 mm (all-air), 6.8–8.2 mm (Region B air only) and 0.95–1.00 mm (all-vacuum). The Region B air contributes $\sigma_B = 6.7\text{--}8.2$ mm in quadrature (3-truth-point mean ≈ 7.4 mm); the Region A air contributes $\sigma_A = 9.8\text{--}12.9$ mm (mean ≈ 11.2 mm).

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1 Geometry and flight path

1.1 Where the target sits

The target centre (tree-gun injection point) is at world coordinates $(-12.5, 0.0, -1069.5)$ mm, tilted by 3° . After the yoke is rotated by -30° about the Y axis, the target's **yoke-local coordinates** are $(-545.5, 0, -919.9)$ mm. The yoke's inner cavity (`vchamber_cavity`) is a $\pm 1620 \times \pm 400 \times \pm 1750$ mm rectangular box plus a downstream trapezoidal extension. Since $z = -919.9$ mm lies inside $[-1750, +1750]$ mm, the target shares a single connected gas/vacuum volume with the magnet cavity.

1.2 The two regions

A particle leaving the target only crosses two materially controllable regions before reaching PDC2:

Table 1: Two-region geometry. Lengths use the straight-line approximation (curvature corrections are second-order and ignored).

#	Physical volume	Length	Switch
A	Magnet cavity <code>vchamber_cavity</code> + <code>VacuumDownstream</code> pipe	~ 4.3 m	<code>/samurai/geometry/BeamLine</code>
B	ExitWindow $\rightarrow$ PDC1 $\rightarrow$ PDC2 (free flight in world volume)	~ 2.0 m	<code>/samurai/geometry/FillAir</code>

In Region A, the straight-line distance from the target to the yoke cavity exit face (yoke-local $z = +1750$ mm) along the local z axis is about 2.67 m (the proton is bent by $\sim 30^\circ$ in the 1.15 T field; arc-length minus straight-line is well under 10%); the pipe segment from the yoke exit to the ExitWindow adds another ~ 1.6 m, totalling ~ 4.3 m. In Region B, ExitWindow $\rightarrow$ PDC1 is about 0.97 m and PDC1 $\rightarrow$ PDC2 is 1.00 m.

1.3 Physical connectivity

The upstream flange of the `VacuumDownstream` pipe attaches to the yoke exit, so geometrically they form a single connected vessel; in code, both volumes' material is selected by the same `BeamLineVacuum` switch: `true` $\rightarrow$ `G4_Galactic` (vacuum), `false` $\rightarrow$ world material (driven by `FillAir`). This is the principal correction in rev2 over stage A', where the pipe was hard-coded as vacuum and decoupled from the magnet cavity — a configuration that is not physically realisable.

1.4 Things to underline

- **Real-experiment configuration is undecided:** whether Region A is air or vacuum in the actual experiment is still under exploration; the three configurations here are simulation ablations, not statements about the planned experimental setup.
- **Target disabled:** all three configurations use `SetTarget false` (tree-gun injection); the target volume itself does not contribute to MS. The material around the target is whatever `vchamber_cavity` contains.

2 The three test configurations

Table 2: The three test configurations. **air**= air (G4_AIR); **vacuum**= vacuum (G4_Galactic).

Configuration	Region A (cavity + pipe)	Region B (EW → PDCs)
all_air	air	air
beamline_vacuum	vacuum	air
all_vacuum	vacuum	vacuum

Geant4 macros (all under configs/simulation/DbeamTest/detailMag1to1.2T/):

Configuration	Key switches
all_air	FillAir true
beamline_vacuum	FillAir true + BeamLineVacuum true
all_vacuum	FillAir false

BeamLineVacuum is independent of FillAir: as long as BeamLineVacuum=true, Region A is vacuum, regardless of FillAir. **all_vacuum** sets FillAir=false so the world volume is also vacuum, and Region A then follows the world material (also vacuum) without any explicit BeamLineVacuum setting.

2.1 Code controls

1. **DipoleConstruction::ConstructSub**: the yoke inner cavity ($vchamber_box \cup vchamber_trap$) is built as its own G4LogicalVolume **vchamber_cavity_log**; with **fBeamLineVacuum=true** its material is G4_Galactic, otherwise the world material.
2. **VacuumDownstreamConstruction::ConstructSub** (*the rev2 change*): the pipe material is also selected by **fBeamLineVacuum**: true → G4_Galactic, false → world material.
3. **DeutDetectorConstruction::Construct**: calls **SetBeamLineVacuum(fBeamLineVacuum)** before instantiating VacuumDownstream; passes the same flag to DipoleConstruction.

3 Configuration and execution

Item	Value
ENSEMBLE_SIZE	50 events per (truth point × configuration)
Seeds	SEED_A=20260421, SEED_B=20260422 (shared across configurations)
Truth points	$(p_x, p_y) = (0, 0), (100, 0), (0, 20)$ MeV/c
Particle	proton, $ \vec{p} = 627$ MeV/c, $\beta_{cp} = 348.4$ MeV
Target	SetTarget false (tree-gun injection)
Reco	test_pdc_target_momentum_e2e --backend rk --rk-fit-mode fixed-target-pdc-only
σ definition	robust half-width $(p_{84} - p_{16})/2$

```
cd /home/tian/workspace/dpol/smsimulator5.5
bash scripts/analysis/ms_ablation/run_ablation.sh
```

Outputs land in build/test_output/ms_ablation_air/: one dispersion_summary.csv per configuration subdirectory plus a top-level compare.csv and three overlay PNGs.

4 Per-configuration σ results

Table 3: PDC hit robust half-width σ (mm) and PDC1→PDC2 slope dispersion σ_θ (mrad). $\sigma_y(\text{PDC2})$ in bold.

Config	p_x	p_y	$\sigma_x(\text{P1})$	$\sigma_y(\text{P1})$	$\sigma_x(\text{P2})$	$\sigma_y(\text{P2})$	σ_{θ_x}	σ_{θ_y}
all_air	0	0	3.071	9.736	3.669	12.826	12.279	9.240
all_air	100	0	3.838	10.492	5.377	12.751	15.768	8.277
all_air	0	20	3.372	13.039	4.065	14.927	21.709	7.846
beamline_vacuum	0	0	1.528	4.061	2.086	6.801	11.834	5.910
beamline_vacuum	100	0	2.067	4.888	3.060	8.211	8.100	5.786
beamline_vacuum	0	20	1.667	4.789	2.363	7.459	14.289	5.421
all_vacuum	0	0	0.147	0.515	0.447	0.947	6.798	2.558
all_vacuum	100	0	0.154	0.364	0.416	0.999	4.075	1.956
all_vacuum	0	20	0.174	0.411	0.396	1.002	6.913	2.299

Across all three truth points every position σ obeys **all_air** > **beamline_vacuum** > **all_vacuum**, consistent with the physical expectation “more air, more MS”. Overlay scatter plots in Figure 1.

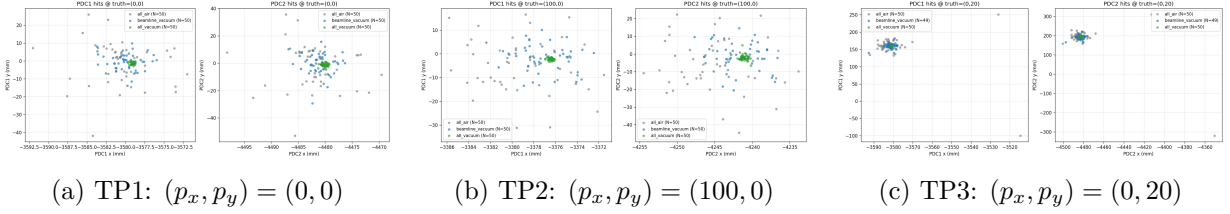


Figure 1: PDC1/PDC2 hit-position scatter for the three configurations ($N = 50$ each). Within each panel, left = PDC1, right = PDC2; horizontal axis x [mm], vertical axis y [mm]. Three colours: **all_air** (grey), **beamline_vacuum** (blue), **all_vacuum** (green).

5 Decomposing the two-region air contribution

5.1 Linear differences

Table 4: Linear differences in $\sigma_y(\text{PDC2})$ (mm).

Difference	Path segment	TP1	TP2	TP3	3-pt mean
all_air – beamline_vacuum	Region A (~ 4.3 m)	6.025	4.540	7.468	6.011
beamline_vacuum – all_vacuum	Region B (~ 2.0 m)	5.854	7.213	6.456	6.508
all_air – all_vacuum	Full air	11.879	11.752	13.925	12.519

The Region A and Region B air contributions to $\sigma_y(\text{PDC2})$ are comparable in magnitude (linear ratio $6.01 : 6.51 \approx 0.92$). Although Region A’s path is more than $2\times$ longer than Region B’s, their contributions are similar because the lever arm projecting Region A’s MS onto the PDC plane (the straight-line drift downstream of the magnet) is shorter than Region B’s — in Region B, the MS happens much closer to the PDCs, so its deflection projects with a larger geometric amplification.

5.2 Quadrature decomposition

$$\sigma_A \equiv \sqrt{\sigma_{\text{all_air}}^2 - \sigma_{\text{bv}}^2}, \quad \sigma_B \equiv \sqrt{\sigma_{\text{bv}}^2 - \sigma_{\text{av}}^2}.$$

Table 5: Quadrature decomposition of $\sigma_y(\text{PDC2})$ (mm). The last two columns verify that $\sqrt{\sigma_A^2 + \sigma_B^2 + \sigma_{\text{av}}^2}$ recovers the measured `all_air` value.

Truth	σ_A (Region A)	σ_B (Region B)	$\sqrt{\sigma_A^2 + \sigma_B^2 + \sigma_{\text{av}}^2}$	$\sigma_{\text{all_air}}$ measured
(0, 0)	10.874	6.735	12.826	12.826
(100, 0)	9.755	8.150	12.751	12.751
(0, 20)	12.930	7.391	14.927	14.927
3-pt mean	11.19	7.43	—	—

The quadrature sum reproduces the measured `all_air` value to floating-point precision (residual $< 10^{-3}$ mm) on all three truth points: the assumption “independent noise sources, σ ’s add in quadrature” is exactly self-consistent on this dataset.

6 Process-noise input for the RK reconstructor

The downstream RK reconstruction (stage D) needs “the MS-induced σ at PDC2 for the chosen experimental configuration”. Depending on whether Region A is going to be evacuated or kept as air in the experiment, two reference values (3-truth-point mean) are tabulated below:

Experimental configuration (Region A state)	$\sigma_{y,\text{MS}}(\text{PDC2})$	$\sigma_{x,\text{MS}}(\text{PDC2})$
A = air (no evacuation), B = air	13.50 mm	4.37 mm
A = vacuum (evacuated), B = air	7.49 mm	2.50 mm

These values are the measured σ ’s of `all_air` and `beamline_vacuum` directly (the residual instrumental term ~ 1 mm from `all_vacuum` is small enough to ignore). Per-truth-point breakdown, Region A evacuated: $\sigma_y(\text{PDC2}) = \{6.80, 8.21, 7.46\}$ mm. The stage D spec should cite the numbers from this memo.

7 Risks and clarifications

- **σ definition:** throughout, robust half-width $(p_{84} - p_{16})/2$. Equal to RMS for a pure Gaussian; conservative by $\sim 10\%$ on heavy-tailed distributions.
- **TP3 `beamline_vacuum`** $N = 49$: one event’s proton failed to match both PDC planes and was naturally excluded by hit matching (2% loss).
- **`all_vacuum` residual** $\sigma_y \sim 1.0$ mm: comes from PDC gas, ExitWindow flange (currently a 30 mm iron flange, not a thin window) and miscellaneous geometry detail; not a target of the decomposition in this memo.
- **Curvature corrections:** the 4.3 m for Region A is along the proton’s curved trajectory; the straight-line approximation differs from the actual arc length by $< 10\%$. We did not validate against Highland (the changing arc length and β make a simple L/X_0 formula imprecise); the quadrature decomposition is taken directly from the measured σ ’s.
- **Target off:** all three configurations use `SetTarget false`, so the proton starts at the target centre.

- **Experimental configuration:** whether Region A will be air or vacuum in the actual experiment is still being discussed; the three configurations here are ablations and are not endorsements of any particular experimental setup.

8 Artefact list

- Scripts: `scripts/analysis/ms_ablation/{run_ablation.sh, compute_metrics.py, compare_ablat.`
- Unit tests: `scripts/analysis/ms_ablation/tests/` (6 tests pass)
- Macros: `configs/simulation/DbeamTest/detailMag1to1.2T/geometry_B115T_pdcOptimized_20260227_target3deg{,_mixed,_noair}.mac`
- Data snapshot: `docs/reports/reconstruction/momentum_reco/ms_ablation/data/{all_air,beamline_vacuum,all_vacuum}_dispersion_summary.csv, compare.csv`
- Figures: `docs/reports/reconstruction/momentum_reco/ms_ablation/figures/dispersion_overlay`

Companion document: `ms_ablation_mixed_20260422_zh.pdf` (Chinese version, identical content).